

A Type Theory beyond Animae

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INTRODUCTION

GOAL: Axiomatising ∞ -category theory within a type theory.

- Riehl and Shulman's "Synthetic Category Theory".
- We focus on a different approach: include **objectwise** statements as opposed to **functorial** ones.

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FUNCTORIALITY: Judgements are "functorial":

$$x : A \vdash f(x) : B(x).$$

(1) This is mapping any $a : A$ into $a \vdash f(a) : B(a)$.

This gives a proof of a predicate about all points in A .

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FUNCTORIALITY: Judgements are "functorial":

$$x : A \vdash f(x) : B(x).$$

- (1) This is not just mapping any $\vdash a : A$ into $\vdash f(a) : B(a)$.
- (2) For any $\alpha : S \rightarrow A$ we can assign $y : S \vdash f(\alpha(y)) : B(\alpha(y))$.

This gives a proof of a predicate about **all maps** into A .

LIMITS OF MARTIN-LÖF TYPE THEORY

MLTT

The rules are the same in **all** contexts.

- If for every $\vdash x : A$ I have $\vdash f(x) : B$, then $x : A \vdash f(x) : B$.
- Pointwise properties of X lift to properties about all maps into X .

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CATEGORY THEORY

- Assignments on objects between categories do not lift to functors.
- Want to access properties over the objects of a category only.

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CATEGORY THEORY

The rules are the same only over **some** contexts.

- Assignments on objects between categories do not lift to functors.
- Want to access properties over the objects of a category only.

ANIMAE AND GROUPOID CORE

STRATEGY: We introduce a class of special contexts: “**animae**”.

- Formulate the theory so that the rules over any $\vdash \Gamma$ anima are always the same as in empty context.
- Given $\vdash \Gamma$ anima, and $\Gamma \vdash A$ type, we have two cases:
 - (1) $\Gamma, x : A$ is an anima $\rightsquigarrow A$ is a “**groupoid**”.
 - (2) $\Gamma, x : A$ is not an anima $\rightsquigarrow A$ is a “**category**”.

If A is a groupoid, we can manipulate it as in MLTT.

Can we “turn” any category into a groupoid?

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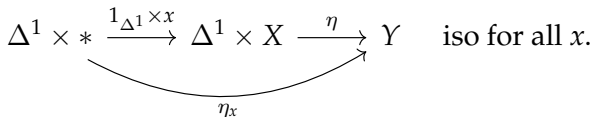
AXIOM: We can assign to any $\vdash A$ type a $\vdash A \simeq \text{gpd}$:

$$\gamma : \Gamma \vdash f(\gamma) : A \quad \rightsquigarrow \quad \gamma : \Gamma \vdash f(\gamma) : A \simeq \quad \text{for any } \vdash \Gamma \text{ gpd.}$$

OBJECTWISE STATEMENTS

We get access to objectwise statements: assume we defined an $\text{interval} \vdash \Delta^1$ type, hence (iso)morphisms in a type.

Ex: Want to characterise $\eta : \Delta^1 \times X \rightarrow Y$ being a natural iso:

$$\Delta^1 \times * \xrightarrow{1_{\Delta^1} \times x} \Delta^1 \times X \xrightarrow{\eta} Y \quad \text{iso for all } x.$$


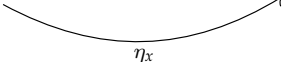
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IN MLTT: we could only write

$x : X \vdash \sigma(x) : \{\eta_x \text{ is an isomorphism}\},$ which means

$$\Delta^1 \times S \xrightarrow{1_{\Delta^1} \times x} \Delta^1 \times X \xrightarrow{\eta} Y \quad \text{iso for all } \vdash S \text{ type and } x.$$


Not the weaker condition we want! Could take $S = X$.

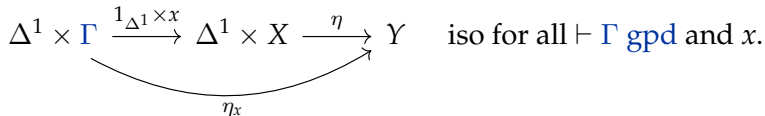
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IN OUR SETTING: The objectwise condition can be written as

$x : X^{\simeq} \vdash \sigma(x) : \{\eta_x \text{ is an isomorphism}\},$ which means

$$\Delta^1 \times \Gamma \xrightarrow{1_{\Delta^1} \times x} \Delta^1 \times X \xrightarrow{\eta} Y \quad \text{iso for all } \vdash \Gamma \text{ gpd and } x.$$


We can finally formulate objectwise conditions in this settings.